DO NOW — RECALL

Trees and binary search trees with arrays

Name: _____ Class: _____ Date: _____

1. What is an array, and how do you access an individual element?

2. What is the purpose of a pointer in a linked list?

3. Explain the difference between LIFO and FIFO. Name one data structure using each.

4. A graph has vertices A, B, C and D. Its undirected edges are A–B, A–C and C–D. Write its adjacency list.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

An array stores indexed elements. Access an element by its index, for example values[2].

2. Model answer

It stores a reference or index identifying another node, allowing nodes to be linked together.

3. Model answer

LIFO: last in, first out (stack). FIFO: first in, first out (queue).

4. Model answer

A: B, C; B: A; C: A, D; D: C. Each undirected edge is represented at both ends.

By the end, you can…

- identify root, parent, child and leaf nodes
- represent a BST using arrays and left/right pointer indexes
- search, add and remove simple BST nodes procedurally

Vocabulary — know these terms

Recall the term — tree

Recall the term — root

Recall the term — parent

Recall the term — child

Recall the term — leaf

Recall the term — binary tree

Recall the term — binary search tree

Recall the term — pointer

NEW LEARNING

Read, annotate and explain

Store left/data/right using arrays

The numbers in brackets are array indexes, not values.

index data left right
0 40 1 2
1 20 3 4
2 60 -1 -1
3 10 -1 -1
4 30 -1 -1

This is the same pointer-table idea used in OCR exam questions: the structure is defined by indexes, not by physical adjacency.

BST ordering

Each comparison chooses the next branch.

Search

Search follows only one branch at each comparison.

Add

The new index is attached to the correct null pointer.

Remove simple cases

Two-child removal is more involved; first secure leaf and one-child cases.

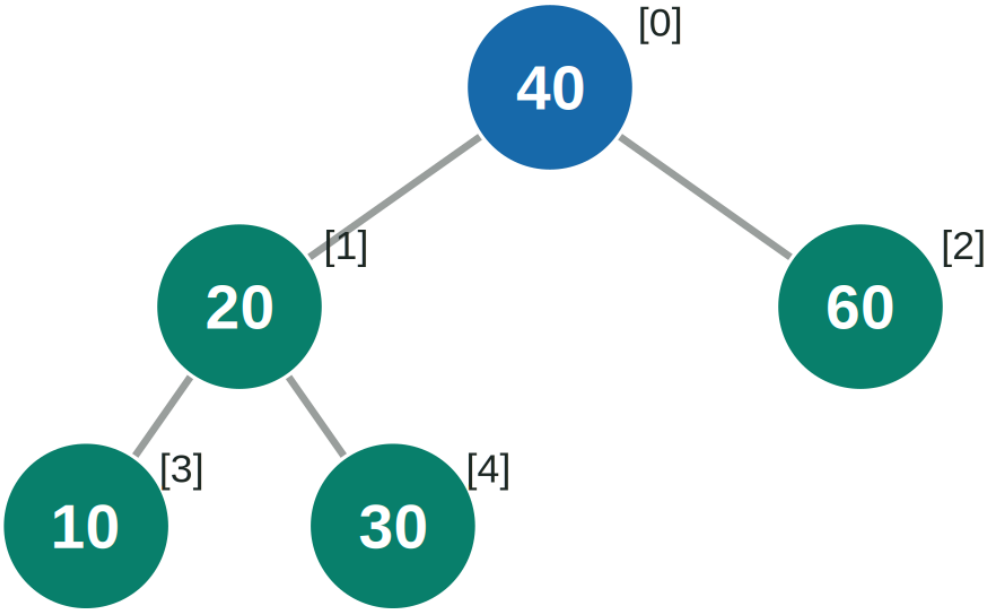
Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Store left/data/right using arrays

The numbers in brackets are array indexes, not values.



Binary search tree: 40 is the root; 20 left and 60 right; 10 and 30 are children of 20

I DO — WORKED EXAMPLE

Follow the worked method

Search for 30

At index 0

```
30 < 40
current = left[0]
current = 1
```

At index 1

```
30 > 20
current = right[1]
current = 4
```

At index 4

```
data[4] = 30
found
```

Skipped nodes

```
Did not inspect indexes 2 or 3.
```

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Represent a BST in a 2D pointer table

OCR uses columns Left / Data / Right.

```
index 0: Sligo
left = ?
right = ?
```

Method / working:

2. Leaf nodes

Cork is a leaf in the OCR tree.

```
index 2: Cork
left = ?
right = ?
```

Method / working:

3. Search path

Search 50 in 40,20,60,50.

```
50 > 40 → ?
50 < 60 → ?
data = 50 → ?
```

Method / working:

4. Remove a leaf

Index 3 is a leaf and left[1] points to 3.

```
left[1] = ?;
```

Method / working:

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Represent a BST in a 2D pointer table OCR uses columns Left / Data / Right. Complete one row CODE: index 0: Sligo left = ? right = ?

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Leaf nodes Cork is a leaf in the OCR tree.

DISCUSS — Compare the representation or steps you used. Explain one decision and challenge one possible misconception.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 3: Search path Search 50 in 40,20,60,50.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 4: Remove a leaf Index 3 is a leaf and left[1] points to 3.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Build a BST with parallel arrays
Represent 40 with children 20 and 60, then 20 with children 10 and 30.

Question 2

Search procedurally
Run the C# starter from I Do. Change the target to 10, 60, then 99. Predict the visit path and result. Then rewrite the search loop yourself.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

June 2024 H446/01 Q8 representation
Using the OCR towns BST, produce the Left/Data/Right table for all eight towns.

Question 4

Remove a leaf using a pointer change
Remove node 10 at index 3 from the example BST, then search for 10.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

In an array-based BST, left and right fields store indexes/pointers, not copies of the child data.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Complete one row

CODE:

index 0: Sligo

left = 1

right = 3

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

Complete the same row

CODE:

index 2: Cork

left = ?

right = ?

CODE:

index 2: Cork

left = null

right = null

IMPROVE: Explain one correction to your own reasoning.

Guided 3 — original CHECK

Complete the path

CODE:

50 > 40 → ?

50 < 60 → ?

data = 50 → ?

CODE:

50 > 40 → right

50 < 60 → left

data = 50 → found

IMPROVE: Explain one correction to your own reasoning.

Guided 4 — original CHECK

Complete the update

CODE:

left[1] = ?;

CODE:

left[1] = -1;

IMPROVE: Explain one correction to your own reasoning.